

Protein Function Prediction

Using Pretrained Language Model Embeddings and Machine Learning

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Motivation and Research Question

The Motivation: Predicting protein function from sequence alone could accelerate research in drug discovery, disease biology, and many other scientific fields.

Research Question: Which embedding and classifier combination best predicts protein Gene Ontology functions, and what do the results reveal about each approach?

Embedding: ProtT5 edged out ESM-2, though the margin is small enough that a strong conclusion cannot be drawn.

Classifier: The Multilayer Perceptron (MLP) outperformed the XGBoost model by 8-10 points. The decisive factor is shared hidden layers; the MLP learns patterns across all 1,585 GO terms simultaneously, while XGBoost does not naturally.

Combination: Stacking three MLPs trained on different embeddings outperforms any single model and creates an overall score competitive with published approaches to this problem.

Related Literature

- **ProtBoost** (Iosipoi & Vakhrushev, NeurIPS 2022)
 - Used ProtT5 + ESM-2 embeddings with Py-Boost gradient boosting and a GCN stacking layer over the GO hierarchy
 - This work informed our classifier design and is the benchmark we most directly compare against.
- **Structure-guided Deep Learning** (Wang et al., Nature Communications, 2025)
 - Combines sequence embeddings with AlphaFold2, a 3-D structure prediction model, to improve protein function annotation
 - Demonstrates the ceiling of what structural information can add on top of sequence embeddings alone

Source: CAFA-6 Protein Function Prediction (Kaggle)

Raw Data

- 82,404 protein sequences
- 537,027 annotation rows mapping proteins to GO terms
- 26,125 unique GO terms total
- Class imbalance is severe

After Preprocessing

- Removed nonstandard amino acid sequences
- Filtered rare GO terms: 26,125 → 1,585 terms
- Final dataset: 76,458 proteins
- Split: 80% train / 20% validation

Three Stage Pipeline: Embed → Classify → Ensemble

Stage 1: Embeddings

- ESM-2 → 1,280-dim
- ProtT5 → 1,024-dim
- Concat → 2,304-dim

Stage 2: Classifiers

- **XGBoost**: one model per GO term
- **MLP**: one network for all 1,585 terms

Stage 3: Ensembles

- MLP + XGBoost average
- **MLP Stack**: average three MLPs across embedding types

10 experiments total, all evaluated using protein-centric $F_{\max}$.

Results

Protein Centric F1 Scoring (Precision and Recall)

Model	Embedding	BP	MF	CC	CAFA Score
MLP	ESM-2	0.2754	0.6587	0.5154	0.4832
MLP	ProtT5	0.2953	0.6561	0.5307	0.4940
MLP	Concat	0.2852	0.6592	0.5236	0.4893
XGBoost	ESM-2	0.2375	0.4841	0.4366	0.3861
XGBoost	ProtT5	0.2510	0.4992	0.4551	0.4018
XGBoost	Concat	0.2505	0.5065	0.4601	0.4057
Ensemble	ESM-2	0.2531	0.5549	0.4751	0.4277
Ensemble	ProtT5	0.2682	0.5643	0.4923	0.4416
Ensemble	Concat	0.2647	0.5693	0.4922	0.4421
MLP	All 3	0.3101	0.6725	0.5363	0.5063

Conclusion

- Both ESM-2 and ProtT5 are capable embedding models for GO function prediction. The gap between them is small, and neither is decisive on its own.
- The bigger lever is the classifier. A neural network that learns across all GO terms simultaneously outperforms any other approach.
- The best result comes not from any single model, but from combining diverse representations of the same data.
- To achieve truly accurate and biologically sound predictions, higher-context modeling is the natural next step in the field.

MultiLayer Perceptron Neural Network

One network predicting all 1,585 GO terms simultaneously.

Layer	Size	Operations
Input	1,280 / 1,024 / 2,304	—
Hidden 1	1,024 neurons	BatchNorm, ReLU, 30% Dropout
Hidden 2	512 neurons	BatchNorm, ReLU, 30% Dropout
Output	1,585 neurons	Sigmoid → probability per GO term

- **Loss:** Focal Loss ($\alpha = 0.25$, $\gamma = 2.0$)
- **Scheduler:** halves learning rate after 3 epochs without improvement; early stopping at 7 epochs
- Output bias initialized to -2.0 to prevent unstable early training given extreme label sparsity

One independent XGBClassifier per GO term. 1,585 models per embedding dataset.

- Each model is a binary classifier: Does this protein have this GO term?
- **Key hyperparameters:** n_estimators=50, max_depth=4, learning_rate=0.3, early_stopping_rounds=5, tree_method=hist, n_jobs=-1.
- scale_pos_weight to handle class imbalance.
- **Critical limitation:** models train independently and cannot share information across related GO terms. The MLP's shared hidden layers solve this; XGBoost cannot.

- T5 encoder-decoder transformer architecture adapted specifically for protein sequences
- Requires space-separated input, each amino acid is its own token: MKTLL $\rightarrow$ M K T L L
- Mean pooling over encoder hidden states produces a single 1,024-dimensional vector per protein
- Outperforms ESM-2 individually on Biological Process and Cellular Component sub-ontologies

- 650 million parameter transformer developed by Meta AI
- Trained on hundreds of millions of protein sequences from the UniRef50 database
- Masked language modeling objective
- Mean pooling across all sequence positions produces a single 1,280-dimensional vector per protein
- Sequences truncated to 1,024 tokens maximum; covers 90.8% of the dataset without truncation

Protein-centric $F_{\max}$. The official CAFA evaluation metric.

- For each threshold $\tau \in [0.01, 0.99]$, binarize predictions: any GO term with predicted probability $\geq \tau$ is called positive
- Compute precision and recall *per protein*, then average across all proteins:

$$\text{precision}(\tau) = \frac{1}{N_p(\tau)} \sum_{i=1}^{N_p(\tau)} \frac{TP_i(\tau)}{P_i(\tau)}, \quad \text{recall}(\tau) = \frac{1}{N} \sum_{i=1}^N \frac{TP_i(\tau)}{T_i}$$

- $F_{\max}$ is the maximum F1 score across all 50 thresholds
- Computed separately for BP, MF, and CC
- Final CAFA score is the arithmetic mean of the three:

$$\text{CAFA Score} = \frac{F_{\max}^{BP} + F_{\max}^{MF} + F_{\max}^{CC}}{3}$$

Sources

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